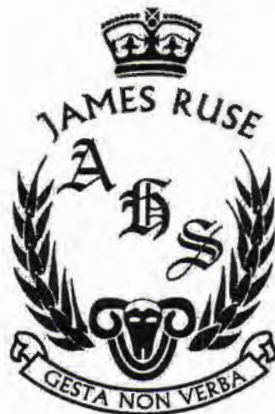


Name:	
Class:	



**YEAR 12
ASSESSMENT TEST 1
TERM 4, 2013**

**MATHEMATICS
EXTENSION 2**

*Time Allowed – 90 Minutes
(Plus 5 minutes Reading Time)*

- All questions may be attempted
- All questions are of equal value
- Department of Education approved calculators and templates are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged work
- No grid paper is to be used unless provided with the examination paper

The answers to all questions are to be returned in separate bundles clearly labeled Question 1, Question 2, etc.

Each question must show (in the top right hand corner) your Candidate Number.

QUESTION 1 (15 Marks)

Marks

- (a) Given that z and w represent the complex numbers $\sqrt{3}+i$ and $1-\sqrt{2}i$ respectively, find:

(i) $\frac{z}{w}$ in the form $x+iy$

2

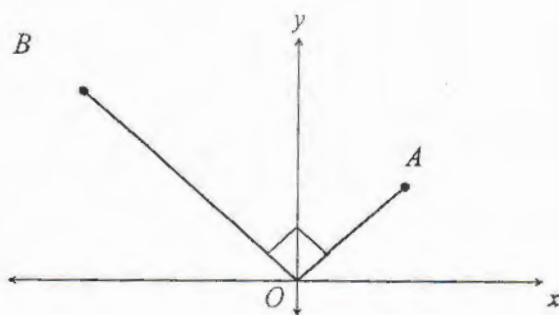
- (ii) the modulus and the argument of z

2

- (iii) the complex number C in $x+iy$ form, given that $\arg C = 2 \arg z$ and $|C| = 3|z|$.

3

- (b)



On the Argand diagram, OA represents the complex number $z_1 = x+iy$, $\angle AOB = \frac{\pi}{2}$ and the length of OB is twice that of OA .

- (i) Find the complex number represented by OB

2

- (ii) Given that $AOBC$ is a rectangle, find the complex number represented by OC .

1

- (iii) Find the complex number represented by BA .

1

- (c) (i) If $(x+iy)^2 = -3-4i$, find all possible values of x and y

2

- (ii) Hence solve the equation $z^2 - 5z + (7+i) = 0$

2

QUESTION 2 (15 Marks)**Start a New page****Marks**

- (a) Indicate the region on the Argand diagram where the following inequalities hold simultaneously

$$2 \leq |z| \leq 4 \quad \text{and} \quad \frac{\pi}{6} \leq \arg z \leq \frac{3\pi}{4}$$

3

- (b) (i) Show that $\frac{(1+i)^8}{(1-\sqrt{3}i)^k} = 2^{4-k} \left[\cos\left(\frac{k\pi}{3}\right) + i \sin\left(\frac{k\pi}{3}\right) \right]$

3

- (ii) For what values of k is $\frac{(1+i)^8}{(1-\sqrt{3}i)^k}$ purely imaginary?

2

- (c) Find the Cartesian equation for the locus of the point z if $\operatorname{Re}\left(z - \frac{1}{z}\right) = 0$

3

- (d) It is known that $5+6i$ is a zero of the polynomial

$$P(x) = 2x^3 - 19x^2 + 112x + d \quad \text{where } d \text{ is real}$$

- (i) What are the other two zeroes of $P(x)$?

2

- (ii) What is the value of d ?

2**QUESTION 3 (15 Marks)****Start a new page****Marks**

- (a) Z lies on the locus defined by $|z+2| = 2$, let $\arg z = \theta$

- (i) By use of an appropriate diagram, show that $\arg(z+2) = 2\theta - \pi$

2

- (ii) Hence, or otherwise, find $\arg(z^2 + 6z + 8)$

2

- (b) The complex number $z=x+iy$ where x and y are real such that $|z-i| = \operatorname{Im}(z)$

- (i) Show that the locus of the point representing z has Cartesian equation $y = \frac{1}{2}(x^2 + 1)$. Sketch this locus

2

- (ii) Find the gradients of the tangents to this curve which pass through the origin. Hence find the set of possible values of the principal argument of z ($-\pi \leq \arg z \leq \pi$)

3

QUESTION 3 (continued)

- | | | | |
|-----|-------|---|---|
| (c) | (i) | If w is one of the complex cube roots of unity, prove that the other complex root is w^2 | 2 |
| | (ii) | Show that $1 + w + w^2 = 0$ | 1 |
| | (iii) | Prove that if n is a positive integer, then $1 + w^n + w^{2n} = 3$ or 0 , depending on whether n is or is not a multiple of 3 | 3 |

QUESTION 4 (15 Marks)

Start a new page

Marks

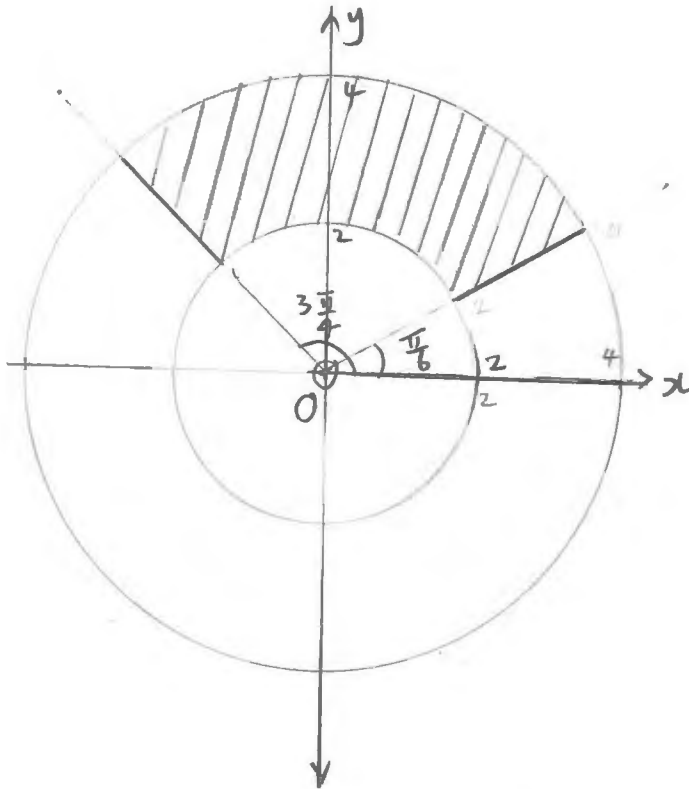
- | | | |
|-----|---|----------------------------|
| (a) | The roots of the equation $z^2 - 6iz + 3 = 0$ are α and β
Without solving this equation. Prove that $ \alpha + \beta \geq 6$ | 2 |
| (b) | <p>(i) Use DeMoivre's Theorem to show that
 $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$ where n is an integer ($\cos \theta \neq 0$)</p> <p>(ii) Use this result to show that the equation
 $(1 + z)^4 + (1 - z)^4 = 0$ has roots of $\pm i \tan \frac{\pi}{8}, \pm i \tan \frac{3\pi}{8}$</p> <p>(iii) Hence or otherwise, show that $\tan^2 \frac{\pi}{8} = 3 - 2\sqrt{2}$</p> | <p>2</p> <p>3</p> <p>2</p> |
| (c) | <p>(i) Find in modulus argument form the roots of the equation $z^{2n+1} = 1$</p> <p>(ii) Hence factorise $z^{2n} + z^{2n-1} + \dots + z^2 + z + 1$ into quadratic factors with real coefficients</p> <p>(iii) Deduce that $2^n \sin \frac{\pi}{2n+1} \sin \frac{2\pi}{2n+1} \dots \sin \frac{n\pi}{2n+1} = \sqrt{2n+1}$</p> | <p>2</p> <p>2</p> <p>2</p> |

END OF EXAM

1	Marks	Comments
(a) (i) $\frac{\sqrt{3}+i}{1-\sqrt{2}i} \times \frac{1+\sqrt{2}i}{1+\sqrt{2}i}$ $= \frac{\sqrt{3}+i+\sqrt{6}i-\sqrt{2}}{1+2}$ $= \frac{\sqrt{3}-\sqrt{2}}{3} + \frac{\sqrt{6}+1}{3}i$	1 1	
(ii) $z = \sqrt{3+1}$ $= 2$ $\text{Arg } z = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ $= \frac{\pi}{6}$	1 1	
(iii) $C = 6$ $\text{Arg } C = 2 \times \frac{\pi}{6}$ $C = 6\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$ $= 6\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$ $= 3 + 3\sqrt{3}i$	1 1 1	
(b) (i) $OB = iOA \times 2$ $= i(x + iy) \times 2$ $= -2y + 2ix$ (ii) $OC = OB + OA$ $= (-2y + 2ix) + (x + iy)$ $= (x - 2y) + (2x + y)i$ (iii) $BA = BO + OA$ $= -OB + OA$ $= -(-2y + 2ix) + (x + iy)$ $= (x + 2y) + (y - 2x)i$	2 1 1	1 for i, 1 for 2

(c) (i) $x^2 - y^2 + 2ixy = -3 - 4i$ Equating terms $x^2 - y^2 = -3$(1) $2xy = -4$(2) $x^2 - \left(\frac{-2}{x}\right)^2 = -3$ $x^4 + 3x^2 - 4 = 0$ $(x^2 + 4)(x^2 - 1) = 0$ $x^2 = 1$ or $x^2 = -4$ $\therefore x = \pm 1$ or no solution When $x=1$, $y=-2$, when $x=-1$, $y=2$	1 1	
(ii) $z = \frac{5 \pm \sqrt{25 - 4(7 + i)}}{2}$ $z = \frac{5 \pm \sqrt{-3 - 4i}}{2}$ $z = \frac{5 \pm (1 - 2i)}{2}$ $z = \frac{6 - 2i}{2}$ or $\frac{4 + 2i}{2}$ $z = 3 - i$ or $2 + i$	1 1	

2a)



1 m for circles
1 m for arg
1 m for shading

b) $1 + i = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$, $1 - \sqrt{3}i = 2 \operatorname{cis}(-\frac{\pi}{3})$

$\frac{1}{2}m + \frac{1}{2}m$

$$\frac{(\sqrt{2} \operatorname{cis} \frac{\pi}{4})^8}{(2 \operatorname{cis} -\frac{\pi}{3})^k} = \frac{2^4 \operatorname{cis}(\frac{8\pi}{4})}{2^k \operatorname{cis}(-\frac{k\pi}{3})} \quad (\text{De Moivre's Th})$$

$\frac{1}{2}k + \frac{1}{2}m$

forgot D.M.Th
- $\frac{1}{2}m$

$$= 2^{4-k} \frac{\operatorname{cis} 2\pi}{\operatorname{cis}(\frac{k\pi}{3})} = 2^{4-k} \operatorname{cis}(2\pi - \frac{k\pi}{3})$$

$\frac{1}{2}m$
(many forgot '---' - $\frac{1}{2}m$)

$$= 2^{4-k} \operatorname{cis}(2\pi + \frac{k\pi}{3}) = 2^{4-k} \operatorname{cis}(\frac{k\pi}{3})$$

$\frac{1}{2}m$ for $2 \operatorname{cis}(2\pi + \frac{k\pi}{3})$

$$= 2^{4-k} (\cos \frac{k\pi}{3} + i \sin \frac{k\pi}{3})$$

or

$$\begin{aligned} \frac{2^{4-k} \operatorname{cis} 2\pi}{\operatorname{cis}(-\frac{k\pi}{3})} &= \frac{2^{4-k} \times 1}{\operatorname{cis}(-\frac{k\pi}{3})} \\ &= 2^{4-k} (\operatorname{cis} -\frac{k\pi}{3})^{-1} \\ &= 2^{4-k} \operatorname{cis} \frac{k\pi}{3} \end{aligned}$$

$\frac{1}{2}m$

$\frac{1}{2}m$

Q 2b. cont'd
 i) $\frac{(1+i)^k}{(1-\sqrt{3}i)^k}$ is purely imaginary if

$$\operatorname{Re} \left(\frac{(1+i)^k}{(1-\sqrt{3}i)^k} \right) = 0$$

$$2. \cos \frac{k\pi}{3} = 0$$

$$\frac{k\pi}{3} = \pm \frac{\pi}{2} + 2n\pi \quad n \in \mathbb{Z}$$

$$k = 6n \pm \frac{3}{2} \quad \#$$

or

$$\cos \frac{k\pi}{3} = \cos \left(\pm \frac{\pi}{2} + n\pi \right)$$

$$\frac{k\pi}{3} = \pm \frac{\pi}{2} + n\pi$$

$$k = \pm \frac{3}{2} + 3n \quad \#$$

$$2c) \quad z - \frac{1}{z} = x+yi - \frac{1}{x+yi} \quad (z \neq 0)$$

$$= x+yi - \frac{1}{x+yi} \times \frac{(x-yi)}{(x-yi)}$$

$$= x+yi - \frac{x-yi}{x^2+y^2}$$

$$\operatorname{Re} \left(z - \frac{1}{z} \right) = 0 \quad \text{if} \quad x - \frac{x}{x^2+y^2} = 0$$

$$x \left[1 - \frac{1}{x^2+y^2} \right] = 0$$

$$\therefore x=0 \quad \text{or} \quad \frac{1}{x^2+y^2} = 1 \quad (x^2+y^2=1)$$

Locus is $x=0$ or $x^2+y^2=1$
 exclude (0,0)

(forgot
 $n \in \mathbb{Z} - \frac{1}{2}\mathbb{Z}$)
 1m

1m

1m

or equivalent

1m

1m

many students wrote $x \neq 0$

(0,0)

2d) Since all coeff of $P(x)$ are real
 $\therefore 5-6i$ is also a root.

Let α be the third root

$$\begin{pmatrix} \text{sum of} \\ \text{Rts} \end{pmatrix} 5+6i + 5-6i + \alpha = \frac{19}{2}$$

$$\therefore \alpha = -\frac{1}{2}$$

So zeros are $5+6i$, $5-6i$ or $-\frac{1}{2}$

$$(ii) \text{ Product of Rts} = -\frac{d}{2}$$

$$(5+6i)(5-6i)\left(-\frac{1}{2}\right) = -\frac{d}{2}$$

$$\therefore d = 25 + 36$$

$$d = 61$$

$$\underline{\underline{61}}$$

Suggested Solutions

Marks

Marker's Comments

$$b) i) \quad z = x + iy \quad x, y \text{ real}$$

$$|z - i| = \text{Im}(z)$$

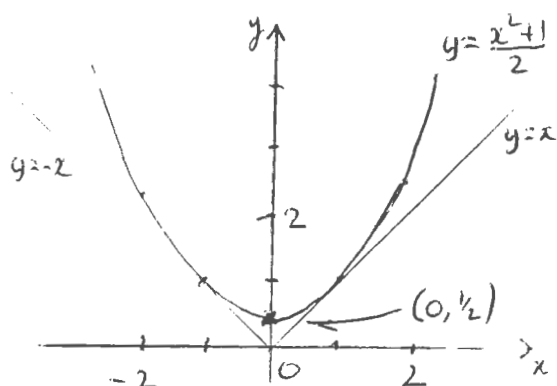
$$|x + i(y - 1)| = y$$

$$x^2 + (y - 1)^2 = y^2$$

$$x^2 + y^2 - 2y + 1 = y^2$$

$$2y = x^2 + 1$$

$$y = \frac{1}{2}(x^2 + 1)$$



ii) Let the tangent through the origin be $y = mx$.

Solve against $y = (x^2 + 1)/2$

$$x^2 + 1 = 2mx$$

$$0 = x^2 - 2mx + 1$$

$$\Delta = 0 \text{ for tangent} \Rightarrow (2m)^2 - 4 = 0$$

$$m = \pm 1$$

All the points z lie between these two tangents which are at angles $\pi/4$ and $3\pi/4$

$$\therefore \underline{\underline{\pi/4 \leq \text{Arg}(z) \leq 3\pi/4}}$$

The diagram needed to show $(0, 1/2)$ as the vertex in some way. It should be an easy mark.

There was some awful garbage presented here.

$\frac{1}{2}$ off for people who got order wrong way.

Suggested Solutions	Marks	Marker's Comments
c) i) $\omega^3 = 1$ (Given - ω is a cube root of unity) $(\omega^3)^3 = \omega^6$ $= (\omega^3)^2 \therefore \omega^2 \text{ is also a root}$ $= 1^2 \quad (\text{but is it different})$ $= 1$ Assume that $\omega^2 = \omega$ Then $\omega(\omega - 1) = 0$ $\therefore \omega = 0$ or 1 since ω is non real. <u>$\therefore \omega^2$ is a different non real root.</u> ii) $1, \omega, \omega^2$ are the three roots of $z^3 - 1 = 0$ Sum of roots = $-\frac{b}{a} = 0$ <u>$\therefore 1 + \omega + \omega^2 = 0$</u> iii) <u>Three cases:</u> <u>n divisible by 3 $n = 3k \quad k \in \mathbb{Z}$</u> $1 + \omega^n + \omega^{2n} = 1 + \omega^{3k} + \omega^{6k}$ $= 1 + (\omega^3)^k + (\omega^3)^{2k}$ $= 1 + 1^k + 1^{2k}$ $= 1 + 1 + 1 = 3$ if not divisible by 3 (a) <u>$n = 3k + 1$</u> $1 + \omega^n + \omega^{2n} = 1 + \omega^{3k} \omega + \omega^{6k} \omega^2$ $= 1 + \omega + \omega^2$ $= 0 \quad (\text{from part ii})$ b) <u>if $n = 3k + 2$</u> $1 + \omega^n + \omega^{2n} = 1 + \omega^{3k} \omega^2 + \omega^{6k} \omega^4$ $= 1 + \omega^2 + \omega^4 = 1 + \omega^2 + \omega$ $= 0$	1 1 1 1 1 1	if $(\omega - 1)(\omega^2 + \omega + 1) = 0$ used then reason should be given why $(\omega - 1) \neq 0$. Lot of waffly answers using patterns etc. Marks were given pro rata.

Conclusion

1/4

MATHEMATICS Extension 2: Question 4

Suggested Solutions

Marks

Marker's Comments

$$(a) \quad z^2 - 6iz + 3 = 0$$

$$\begin{aligned} \text{sum of roots: } \alpha + \beta &= -b/a \\ &= -(-6i) \\ &= 6i \end{aligned}$$

1 mk

$$|\alpha| + |\beta| \geq |\alpha + \beta| \quad \text{by the triangle inequality}$$

$$\therefore |\alpha| + |\beta| \geq 6$$

-1/2 mk if you didn't mention the triangle inequality

1

$$(b)(i) \quad \text{LHS} = (1 + i \tan \theta)^n + (1 - i \tan \theta)^n$$

$$= \left(\frac{\cos \theta + i \sin \theta}{\cos \theta} \right)^n + \left(\frac{\cos \theta - i \sin \theta}{\cos \theta} \right)^n$$

$$= \frac{\cos^n \theta + i \sin^n \theta}{\cos^n \theta} + \frac{\cos^n \theta - i \sin^n \theta}{\cos^n \theta}$$

$$= \frac{\cos^n \theta + i \sin^n \theta + \cos^n \theta - i \sin^n \theta}{\cos^n \theta}$$

$$= \frac{2 \cos^n \theta}{\cos^n \theta}$$

$$= \text{RHS}$$

by de Moivre's Theorem

1 mk

$$(ii) \quad (1+z)^4 + (1-z)^4 = 0$$

$$\text{let } z = i \tan \theta$$

$$(1 + i \tan \theta)^4 + (1 - i \tan \theta)^4 = 0$$

$$\frac{2 \cos^4 \theta}{\cos^4 \theta} = 0 \quad \text{from (i)}$$

$$\therefore 2 \cos^4 \theta = 0$$

$$\cos^4 \theta = 0$$

1 mk

* Trig equation so $4\theta = \pm \pi/2, \pm 3\pi/2$ (as x^4 is a quartic and so only has 4 solns)

it could be done using general solutions but most forget the formulae

$$\theta = \pm \pi/8, \pm 3\pi/8$$

$$\text{now } z = i \tan \theta$$

$$\therefore z = i \tan \pi/8, i \tan -\pi/8, i \tan 3\pi/8, i \tan -3\pi/8$$

1 mk

$$\therefore z = i \tan \pi/8, -i \tan \pi/8, i \tan 3\pi/8, -i \tan 3\pi/8 \quad (\text{as } \tan -\theta = -\tan \theta)$$

* max. 2 mks

* most students didn't explain why they had 4 solutions they appear to just "copy" what was given in the question!

MATHEMATICS Extension 2: Question 4.....

Suggested Solutions	Marks	Marker's Comments
(b)(iii) $(1+z)^4 + (1-z)^4 = 0$ $1 + 4z + 6z^2 + 4z^3 + z^4 + 1 - 4z + 6z^2 - 4z^3 + z^4 = 0$ $2 + 12z^2 + 2z^4 = 0$ $1 + 6z^2 + z^4 = 0$ $z^2 = \frac{-6 \pm \sqrt{32}}{2}$ $= -3 \pm \sqrt{8}$ $= -3 \pm 2\sqrt{2}$ now $z = i \tan^{\pi/8}$ $(i \tan^{\pi/8})^2 = -3 \pm 2\sqrt{2}$ $-1 \tan^2 \pi/8 = -3 \pm 2\sqrt{2}$ $\tan^2 \pi/8 = 3 \mp 2\sqrt{2}$ now $0 < \tan^{\pi/8} < 1$ (as $\tan^{\pi/4} = 1$) $\therefore 0 < \tan^2 \pi/8 < 1$ $\therefore \tan^2 \pi/8 = 3 - 2\sqrt{2}$ * needed to justify properly why it was $3 - 2\sqrt{2}$, a lot of students fudged this!!!	1mk Quadratic formula 1/2 mk only	1/2 mk.

3/4

MATHEMATICS Extension 2: Question....

4

Suggested Solutions

Marks

Marker's Comment

Question 4 $z^{2n+1} - 1 = 0$

(c) (i) $z^{2n+1} = 1$

let $z = \text{cis } \theta$

$(\text{cis } \theta)^{2n+1} = 1$

(by De Moivre's)

$\text{cis}(2n+1)\theta = \text{cis}(2k\pi)$ where

$(2n+1)\theta = 2k\pi$

$\theta = \frac{2k\pi}{2n+1}$ for $k=0, 1, 2, \dots, 2n$
or $k=0, \pm 1, \pm 2, \dots, \pm n$

$\therefore z = \text{cis } 0, \text{cis } \frac{2\pi}{2n+1}, \text{cis } \frac{4\pi}{2n+1}, \dots, \text{cis } \frac{2n\pi}{2n+1}$

OR $\text{cis } 0, \text{cis } \frac{2\pi}{2n+1}, \text{cis } \frac{-2\pi}{2n+1}, \dots$

(ii) $z^{2n+1} - 1 = (z-1)(z^{2n} + z^{2n-1} + \dots + z + 1)$

SO $z^{2n} + z^{2n-1} + \dots + z + 1$

$= (z - \text{cis } \frac{2\pi}{2n+1})(z - \text{cis } \frac{4\pi}{2n+1}) \dots (z - \text{cis } \frac{2n\pi}{2n+1})$
 $(z - \text{cis } \frac{2\pi}{2n+1})(z - \text{cis } \frac{-2\pi}{2n+1}) \dots (z - \text{cis } \frac{2n\pi}{2n+1})$

Equating conjugate pairs

$(z - \text{cis } \frac{2\pi}{2n+1})(z - \overline{\text{cis } \frac{2\pi}{2n+1}}) \dots (z - \text{cis } \frac{2n\pi}{2n+1})(z - \overline{\text{cis } \frac{2n\pi}{2n+1}})$

$\text{cis } \frac{2\pi}{2n+1} \text{ and } \overline{\text{cis } \frac{2\pi}{2n+1}}$

Students

using $\frac{2n\pi}{2n+1}$

got 0.

needed to
indicate
end points
 $k=0, \pm 1, \pm 2, \dots, \pm n$ or poorly done
too many should
did not indicate
values for k.

$z^{2n+1} = 1 \quad z = \text{cis } \frac{2k\pi}{2n+1}$

$k = 0, \pm 1, \pm 2$

MATHEMATICS Extension 2: Question 4...

Suggested Solutions	Marks	Marker's Comment
Question 4 (c) $\therefore (z^2 - 2\cos\frac{2\pi}{2n+1}z + 1)(z^2 - 2\cos\frac{4\pi}{2n+1}z + 1) \dots (z^2 - 2\cos\frac{2n\pi}{2n+1}z + 1) - \textcircled{1}$ needed to state and prove (ii) Now $z^{2n} + z^{2n-1} + \dots + z^2 + 1 =$ $(z^2 - 2\cos\frac{2\pi}{2n+1}z + 1) \dots (z^2 - 2\cos\frac{2n\pi}{2n+1}z + 1)$ Let $z = 1$ $1 + 1 + \dots + 1 + 1$ $= (1 - 2\cos\frac{2\pi}{2n+1} + 1)(1 - 2\cos\frac{4\pi}{2n+1} + 1) \dots (1 - 2\cos\frac{2n\pi}{2n+1} + 1) - \textcircled{1}$ if $2n+1$ is odd, then $2n+1 = 2(1 - \cos\frac{2\pi}{2n+1}) \dots 2(1 - \cos\frac{2n\pi}{2n+1})$ Now $1 - \cos 2\theta = 2\sin^2 \theta$ needed to state and prove $2n+1 = 2^n \cdot \left(2\sin^2\frac{\pi}{2n+1} \cdot 2\sin^2\frac{2\pi}{2n+1} \dots 2\sin^2\frac{n\pi}{2n+1} \right)$ $2n+1 = 2^{2n} \left(\sin^2\frac{\pi}{2n+1} \sin^2\frac{2\pi}{2n+1} \dots \sin^2\frac{n\pi}{2n+1} \right) \text{ problems with power of 2.}$ $\sqrt{2n+1} = 2^n \sin\frac{\pi}{2n+1} \sin\frac{2\pi}{2n+1} \dots \sin\frac{n\pi}{2n+1} - \textcircled{1}$		